

REXMI - Physical AI for space

The first robot

A wheeled quadruped designed for lunar surface work, with crater access built into the system concept. Wheels cover traversable ground; legs adapt clearance and contact placement across broken terrain. Reinforcement learning and onboard navigation connect movement to mission execution.

Reach the site. Collect the evidence. Return useful data.

Mission roles

Different roles on a common mobility platform, configured around the mission and payload.

- **Scout and recon.** Map routes, assess terrain and locate points of interest.
- **Crater access and science.** Reach selected crater rims, slopes and interior sites to support local scientific measurements and resource characterization.
- **Supplies.** Carry mission-sized instruments, equipment and small loads between surface locations.
- **Inspection.** Revisit infrastructure and instruments to document condition and changes.

Why crater access matters

Permanently shadowed regions in polar craters can preserve water ice and other volatiles. Crater walls and ejecta also expose material useful for understanding lunar geology. Access lets a mobile instrument investigate promising sites identified from orbit. [1, 2]

The opportunity extends beyond craters: water also occurs on sunlit terrain, and regolith contains minerals bearing oxygen, iron, titanium and other elements. Resource value depends on local concentration, accessibility and extraction conditions. [1, 2]

From orbital clues to ground measurements

REXMI's planned science role is to navigate to a selected point, acquire measurements from integrated instruments and return location-tagged data. Candidate payloads include spectrometers for composition or volatile signatures and ground-penetrating radar (GPR) for subsurface structure. Local observations can test and refine interpretations from orbit. [3, 4]

GPR reveals subsurface contrasts; identifying water or specific elements requires complementary measurements. The instrument suite will be selected for the scientific question.

Current foundation

REXMI's development includes reinforcement-learning mobility, navigation, movement-skill switching and recovery in simulation. Crater access is a current design objective. Physical hardware validation and lunar environmental qualification are planned development work. [5]

Development plans

Five connected workstreams support the mission roles.

Lunar mobility simulation

Planned training and evaluation cover reduced gravity, loose terrain, traction variation and actuator limits. Crater scenarios include approach, descent, traverse and exit, with stability, route completion, energy use and recovery as evaluation measures.

Adaptive thermal control

The thermal architecture will combine heat paths, radiators, insulation, heaters and temperature-aware operation. Planned analysis links payload and actuator heat to terrain, illumination and mission duration, including the demands of shadowed crater operations.

Perception

Planned onboard sensing and localization will support terrain mapping, hazard detection, traversability assessment and precise positioning at science targets. Instrument observations will be associated with location, time and acquisition conditions for scientific interpretation.

Dust protection

Mechanism development will address shields, cable routing, bearing seals and lubrication, alongside protection for optics and radiators. Planned testing covers dust exposure, wear, vacuum and thermal cycling.

Robot validation

Planned hardware trials will evaluate mobility, instrument operation, load carrying and inspection on representative terrain. Validation will measure repeatability, data quality, energy use and operator intervention, with protective responses to lost localization, communication faults and thermal limits.

Mission and payload integration

The system plan includes payload mounting, power and data interfaces, instrument calibration, communications and onboard data storage. Mission design will define mass, energy reserves, thermal limits and return paths. Access to permanent shadow requires its own power, thermal and communications provisions.

Long-term vision: physical intelligence for space, across quadrupeds, humanoids and new hybrids.

Sources

1. [NASA: Moon Water and Ices](#). Polar ice and water on sunlit terrain.
2. [NASA: Moon Composition](#). Lunar minerals, regolith and impact-excavated material.
3. [NASA: Lunar Terrain Vehicle instruments](#), 10 July 2025. Spectroscopy and GPR science roles.
4. [NASA: Lunar surface science payloads](#). Local measurements to ground-truth orbital observations.
5. REXMI project descriptions, August 2026; founder design direction, September 2026. Development status and plans.